Operating systems concepts lab experiments

Ex : 4 system calls

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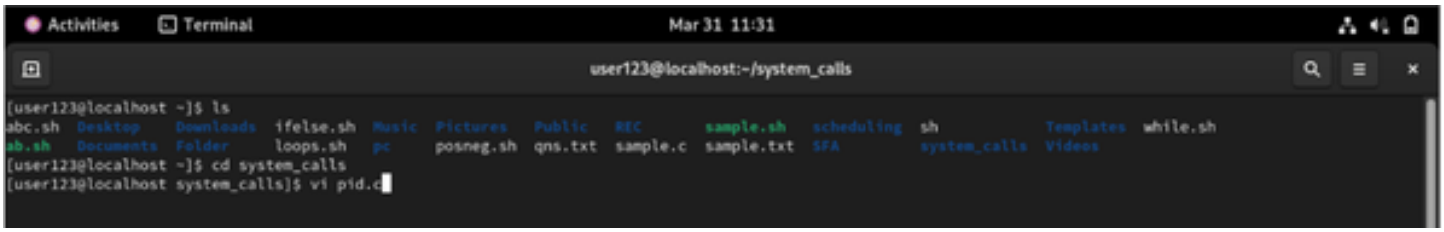

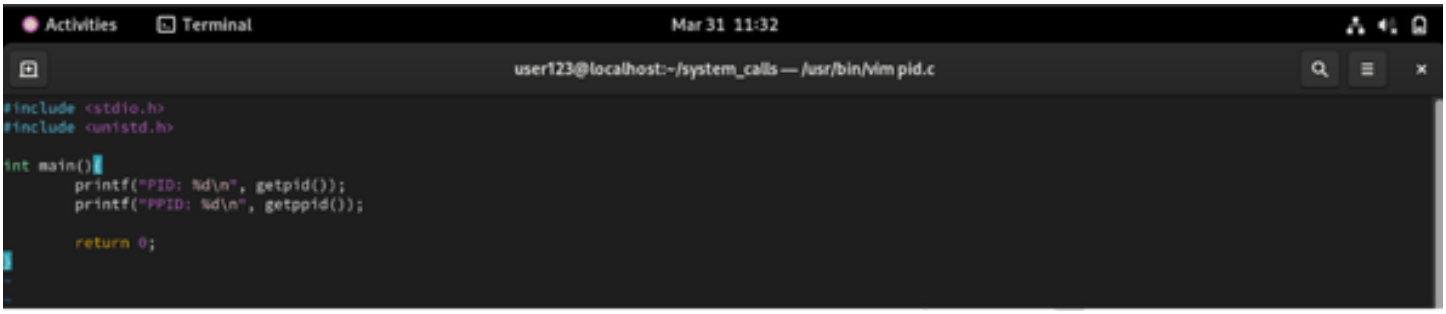

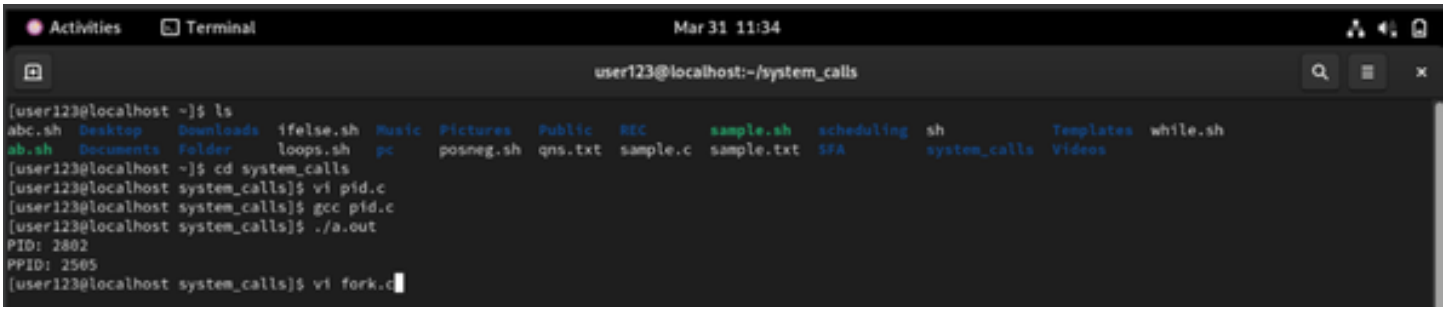

Activities Terminal Mar 31 11:35

user123@localhost:~/system_calls — /usr/bin/vim fork.c

```c
#include <stdio.h>
#include <unistd.h>

int main(){
        if(fork()==0){
                printf("This is CHILD process\n");
        } else {
                printf("This is PARENT process\n");
        }

        printf("PID: %d\n", getpid());

        return 0;
}
```

Activities Terminal Mar 31 11:37

user123@localhost:~/system_calls

```
[user123@localhost ~]$ ls
abc.sh  Desktop    Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh   scheduling  sh           Templates  while.sh
ab.sh   Documents  Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  SFA         system_calls  Videos
[user123@localhost ~]$ cd system_calls
[user123@localhost system_calls]$ vi pid.c
[user123@localhost system_calls]$ gcc pid.c
[user123@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[user123@localhost system_calls]$ vi fork.c
[user123@localhost system_calls]$ gcc fork.c
[user123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[user123@localhost system_calls]$ vi exec.c
```

Activities Terminal Mar 31 11:39

user123@localhost:~/system_calls — /usr/bin/vim exec.c

```c
#include <stdio.h>
#include <unistd.h>

int main(){
        printf("Before exec()\n");
        execl("/bin/ls", "ls", NULL);
        printf("After exec()\n");

        return 0;
}
```

Activities Terminal Mar 31 11:41

user123@localhost:~/system_calls

```
[user123@localhost ~]$ ls
abc.sh  Desktop   Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh   scheduling  sh           Templates  while.sh
ab.sh   Documents Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  SFA         system_calls Videos
[user123@localhost ~]$ cd system_calls
[user123@localhost system_calls]$ vi pid.c
[user123@localhost system_calls]$ gcc pid.c
[user123@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[user123@localhost system_calls]$ vi fork.c
[user123@localhost system_calls]$ gcc fork.c
[user123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[user123@localhost system_calls]$ vi exec.c
[user123@localhost system_calls]$ gcc exec.c
[user123@localhost system_calls]$ ./a.out
Before exec()
a.out  exec.c  fork.c  opendir.c  pid.c  readdir.c
[user123@localhost system_calls]$ vi opendir.c
```

Activate Windows

Activities Terminal Mar 31 11:41

user123@localhost:~/system_calls — /usr/bin/vim opendir.c

```
#include <stdio.h>
#include <dirent.h>

int main(){
        DIR *dir;

        dir = opendir(".");

        if(dir == NULL){
                printf("Failed to open directory\n");
                return 1;
        }

        printf("Directory opened successfully!\n");
        closedir(dir);
        return 0;
}
~
~
~
~
```

Activities Terminal Mar 31 11:42

user123@localhost:~/system_calls

```
[user123@localhost ~]$ ls
abc.sh  Desktop   Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh   scheduling  sh           Templates  while.sh
ab.sh   Documents Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  SFA         system_calls Videos
[user123@localhost ~]$ cd system_calls
[user123@localhost system_calls]$ vi pid.c
[user123@localhost system_calls]$ gcc pid.c
[user123@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[user123@localhost system_calls]$ vi fork.c
[user123@localhost system_calls]$ gcc fork.c
[user123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[user123@localhost system_calls]$ vi exec.c
[user123@localhost system_calls]$ gcc exec.c
[user123@localhost system_calls]$ ./a.out
Before exec()
a.out  exec.c  fork.c  opendir.c  pid.c  readdir.c
[user123@localhost system_calls]$ vi opendir.c
[user123@localhost system_calls]$ gcc opendir.c
[user123@localhost system_calls]$ ./a.out
Directory opened successfully!
[user123@localhost system_calls]$ vi readdir.c
```

Activities Terminal Mar 31 11:44

user123@localhost:~/system_calls — /usr/bin/vim readdir.c

```c
#include <stdio.h>
#include <dirent.h>

int main(){
        DIR *dir;
        struct dirent *entry;

        dir = opendir(".");

        if(dir == NULL){
                printf("Cannot open directory\n");
                return 1;
        }

        printf("Files in directory:\n");

        while((entry=readdir(dir)) != NULL){
                printf("%s\n", entry->d_name);
        }

        closedir(dir);

        return 0;
}
```

Activities Terminal Mar 31 11:45

user123@localhost:~/system_calls

```
[user123@localhost ~]$ ls
abc.sh  Desktop   Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh   scheduling  sh           Templates  while.sh
mb.sh   Documents Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  SFA         system_calls Videos
[user123@localhost ~]$ cd system_calls
[user123@localhost system_calls]$ vi pid.c
[user123@localhost system_calls]$ gcc pid.c
[user123@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[user123@localhost system_calls]$ vi fork.c
[user123@localhost system_calls]$ gcc fork.c
[user123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[user123@localhost system_calls]$ vi exec.c
[user123@localhost system_calls]$ gcc exec.c
[user123@localhost system_calls]$ ./a.out
Before exec()
a.out  exec.c  fork.c  opendir.c  pid.c  readdir.c
[user123@localhost system_calls]$ vi opendir.c
[user123@localhost system_calls]$ gcc opendir.c
[user123@localhost system_calls]$ ./a.out
Directory opened successfully!
[user123@localhost system_calls]$ vi readdir.c
[user123@localhost system_calls]$ gcc readdir.c
[user123@localhost system_calls]$ ./a.out
Files in directory:
.
..
pid.c
fork.c
exec.c
opendir.c
readdir.c
a.out
[user123@localhost system_calls]$
```

Activate Windows
Go to Settings to activate W